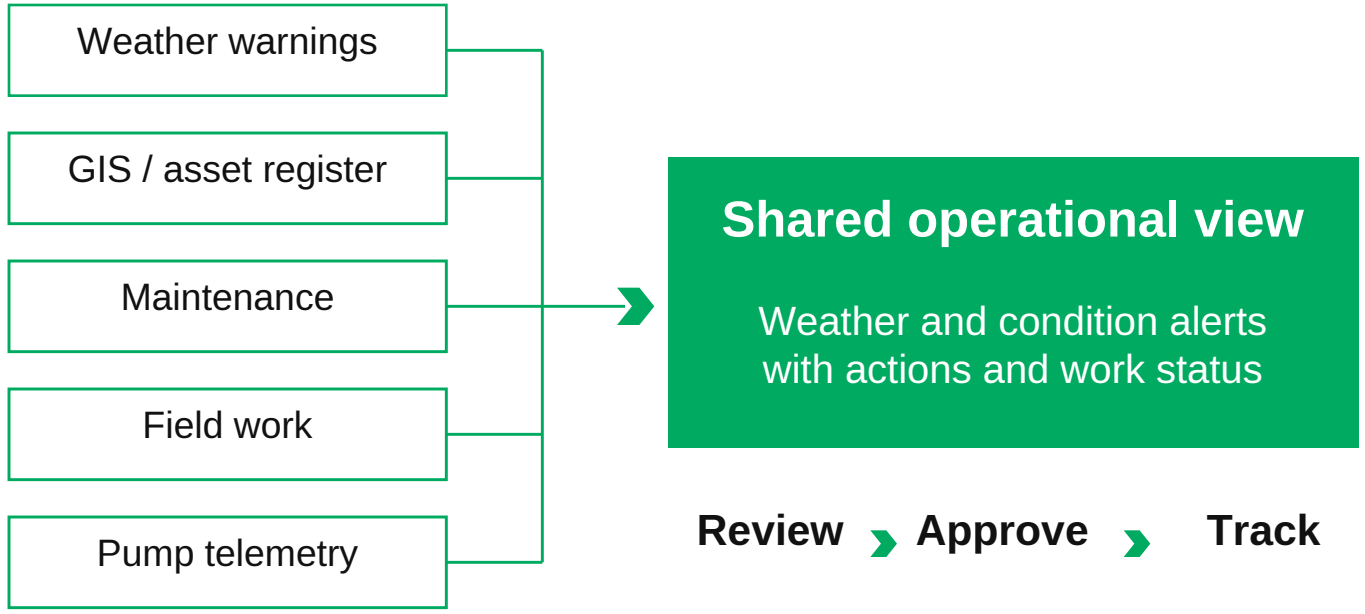
SGW Resilience - Proposed Solution

Flood preparedness and pump condition review through one shared response workflow.
Initial pilot: flood readiness for selected pumping stations and substations.

THE OPERATIONAL PROBLEM

Fragmented data makes it difficult to assess asset exposure, equipment condition and response priorities.



AI SUPPORT

AI flags unusual pump behaviour, drafts actions with supporting sources and prepares leadership briefings.

Prototype uses simulated data and procedures.

Strategic Business Value

Improve preparedness and response to help protect essential services.

SGW PRIORITY	EXPECTED OPERATIONAL VALUE	MEASURE IN THE PILOT
✓ **Prioritize assets at risk**	Focus review and response on critical assets requiring attention.	Prioritization time and engineer assessment
✓ **Coordinate response**	Reduce gaps and duplicated work through clear ownership.	Unassigned and duplicate tasks
✓ **Support leadership decisions**	Provide timely, consistent updates on priorities and unresolved work.	Briefing preparation time and factual accuracy

Expected benefits, not measured client results. Baselines and targets will be agreed with SGW.

Financial Implications & ROI

Expected return: staff capacity released for priority work, with cost savings where validated.

COST DRIVERS

Implementation

Development, integration, data preparation, security and user training.

Operation and support

Hosting, AI usage, monitoring and ongoing support.

EFFICIENCY & RETURN

- **Estimate the value of staff time saved.**

  Less manual data gathering and briefing preparation.

- **Confirm actual cost savings.**

  Check whether overtime or external support costs can be reduced.

- **Compare benefits and total costs over the same period.**

  Include both implementation and ongoing operating costs.

Time released is staff capacity; cash savings require validation with SGW finance.

Delivery Roadmap

Proposed pilot: one region, selected pumps and substations, flood readiness.
Pump monitoring subject to telemetry readiness and validation.

	PHASE	MAIN WORK	MILESTONE
1	**Understand needs**	Agree scope, data, operating procedures and success measures.	Pilot scope, data access and success measures agreed.
2	**Prepare the pilot**	Connect sources, establish access controls and test the workflow.	Workflow and security checks passed.
3	**Test with users**	Train teams and validate the workflow in operational scenarios.	Pilot results reviewed with operations.
4	**Decide next steps**	Review business value, cost and remaining risks.	Decision agreed: expand, improve or stop.

Key dependencies: data access and quality, agreed procedures and operational team availability.

Dates and budget will be confirmed after reviewing scope and data readiness.

Governance & Compliance

Manage operational risk through clear accountability and controlled AI use.

CONTROLS REQUIRED BEFORE OPERATIONAL USE

- ✓ **Accountability** — Named decision owners, authorized users and human approval.
- ✓ **Data protection** — Agreed data access, retention and use of external AI services.
- ✓ **AI assurance** — Traceable decisions, validated AI outputs and ongoing monitoring.

Authorized people approve operational actions.

No equipment control or automatic dispatch.

SGW security, legal and operations teams must confirm applicable requirements and approve the controls.

Scalability & Next Steps

Expand after pilot validation, adapting the workflow to each operating context.

EXPAND THE PLATFORM

Regions

Extend to more SGW regions.

Hazards

Add hurricanes, heatwaves and wildfires.

Infrastructure

Add water treatment and transmission.

FUTURE AI OPPORTUNITIES

Optimization

FIRST CANDIDATE

Support crew allocation and work scheduling.

Forecasting

Estimate response workload to support resource planning.

Computer Vision

Support damage screening from inspection images.

Reuse the core workflow; validate data and operating rules for each expansion.

Recommended next step: agree the pilot scope, owners and success measures with SGW.

Expansion depends on measured value, user adoption and acceptable operational risk.